# TFPT — Frontier items

$\eta_B$, $m_p/m_e$, the Koide relation, dark matter and quantum gravity — the honest status, not forced onto the ladder

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**What this document is about — the status authority for the frontier**

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of $\eta_B$, $m_p/m_e$, the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

**The TFPT document set — what is treated where**

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point $\alpha^{-1}$, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one $\varphi_0$-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, $Q_+$ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven $E_8$ slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of $\eta_B$, $m_p/m_e$, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions: R.** `tfpt_research_contracts` — the open interfaces $(v_{\mathrm{geo}}, G_{\mathrm{net}}, F_{\mathrm{transfer}})$ as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

**You are reading document 4 (Frontier).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\mathrm{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface | Question | Status |
|---|---|---|
| $v_{\mathrm{geo}}$ | the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale | primitive [O] |
| $G_{\mathrm{net}}$ | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1, 1)\rangle \cong (E_8)_1$ | [C] |
| $F_{\mathrm{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$, axion, $m_p/m_e$) | [C] |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so $c_3$ is triply over-determined — anchor, geometry, thermodynamics, `v358`) and $\Lambda$ from $\alpha$ ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit $v_{\mathrm{geo}}$; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem $G_{\mathrm{net}}$ (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) is now *closed modulo a cited theorem*. The explicit lattice model (`v367`/`v368`) and the $S_3$ closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ($h_s$=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM $\mu$=4/2$^2$=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so this is *closed modulo a cited theorem*, not solved. `QGEO.SYM.01` (the $\mu_4$ deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76`/`v330`). So the honest residual is the one unit $v_{\mathrm{geo}}$ plus the typed $F_{\mathrm{transfer}}$ interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\mathrm{wall}}, G_{\mathrm{metric}}, F_{\mathrm{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\mathrm{geo}} \oplus F_{\mathrm{transfer}}$, with the seam $G_{\mathrm{net}}$ closed modulo a cited theorem and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven   [C] conditional   [O] open / axiom   [X] kill test

**P1** $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8,\ h = 30,\ \varphi_0$ *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix $R$ *([E])*

↓

Physical bridges: $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ *([C] bridges)*

↓

Frontier readouts: $\Lambda$, inflation, $\eta_B$, dark matter, QG *([C]/[O])*

# Contents



**Ground rule.** These five are exactly the items previously flagged "*not on a clean action/ladder yet — do not force.*" This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers [E]/[C]/[O] are applied throughout. (All numbers machine-verified.)

**Hard rule on the frontier — one interface $F_{\text{transfer}}$**

**Koide, $\eta_B$, the axion relic scale and $m_p/m_e$ are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied.** They are deliberately typed interfaces, not structural holes, and they are *four instances of one missing functor* $F_{\text{transfer}}$ — the continuous transport from compiler *source* data to measured observables (Koide source→pole; $\eta_B$ source→Boltzmann relic; axion scale→cosmological abundance; $m_p/m_e$ → QCD/EW matching). Each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on that named transfer calculation — now a runnable [C] solver with a kill test (v371–v375), still a transfer target, never a compiler power. In particular $\eta_B$ is a *transfer target*, not a primitive compiler prediction. They must never be quoted as primitive TFPT outputs.

*Why this is principled, not a gap:* these are precisely the **discrete↔ continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [verification/v35_quark_qcd_boundary.py]

*Machine-enforced.* This rule is a CI-style guard ([verification/v187_ftransfer_laws.py]): the four frontier families ($F_{\text{pole}}$ Koide, $F_{\text{Boltzmann}}$ $\eta_B$, $F_{\text{relic}}$ axion, $F_{\text{QCD}}$ $m_p/m_e$) are read from the ledger and required to carry a [C]/[O] marker — exact algebraic sub-parts (the Koide 53/54 factor, $b_3 = -7$) may be [E], but the *physical* prediction never is. No future edit can quietly promote a frontier number to a closed compiler output. A companion *prose* sentinel ([verification/v188_frontier_wording_guard.py]) forbids stale sentences whose semantics an earlier round superseded, so a frozen release can never re-introduce wording that was true yesterday and is half-true today.

*The functor contract* `CONTRACT.F.01`*.* Beyond the typing guard, the functor is pinned by four checkable structural axioms ([verification/v213_ftransfer_functor.py]): each instance is a *discrete* compiler kernel [E] composed with an *external* continuous solver [C], and the functor is (1) $\mu_4$-deck equivariant (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ *is* the deck transfer eigenvalue), (2) Plücker-preserving ($53 = a^T(R+Q)\mathbf{1}$, $\|\text{Pl}(K)\|_1 = 11$), (3) positive/stochastic ($\operatorname{spec} T = \{1, (2/3)^6, (1/3)^6\}$, $\kappa_f \in (0,1)$), and (4) external-module-explicit ($b_3 = -7$ a carrier output, the Boltzmann/relic/QCD content named and external). So $F_{\text{transfer}}$ is a typed functor, not a bag of open topics — the third research contract alongside ($U_{\text{wall}}$) and ($G_{\text{metric}}$), and a consolidation, not a closure.

*How external is "external"? A falsifiable clock probe ([verification/v397_external_clock_probe.py]).* Under the strict No-Free-Pattern rule (an *exact* match, not a wide-band fit), three of the four frontier scheme-numbers in fact land *on* the $\{2,3,5\}$ clock: the $\eta_B$ decuple $A_\Lambda = 10 = |Z_2|\, g_{\text{car}}$ (v212), the Koide rate $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$ (v303) and the axion spine angle $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (v211). Only the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value (several simple numbers fit its $\pm 5.3\%$ band), so it stays genuinely external — the irreducible "external physics" residue shrinks to essentially the proton's non-perturbative QCD number.

*The Paper-E closure certificate ([verification/v402_ftransfer_suite.py],* `FTRANSFER.SUITE.01`*).* The four sectors are certified as ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$: each observable factors source → threshold → RG → observable from an exact atom source ($Q_{\text{src}}$, $A_\Lambda = 2g_{\text{car}} = 10$, $f_a = M_{\text{scal}}/128$, $b_3 = -7$) and carries a committed kill test, with the firewall intact (the physical prediction stays [C]/[O], only the algebraic source [E]) — a consolidation of the `FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE` solvers, not a closure.

# 1 Baryon asymmetry $\eta_B$ — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

**Route A — from the closed baryon fraction.** The seed already fixes the baryon *fraction* (the $E_8$ decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \tfrac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \qquad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed $\Omega_b \approx 0.049$. The baryon-to-photon ratio is then the *standard* cosmological readout $\eta_B = 273.9 \times 10^{-10}\,\Omega_b h^2$. With the TFPT Hubble value ($h \approx 0.674$, from $H_0 \sim \sqrt{\Lambda}$),

$$\boxed{\Omega_b h^2 = 0.0222, \qquad \eta_B = 6.09 \times 10^{-10}} \qquad (\text{observed } 6.1 \times 10^{-10}).$$

So $\eta_B$ is a *downstream cosmological readout* from the closed $\Omega_b h^2$ handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

**Route B — leptogenesis interface ([C]).** The TFPT branch fixes the neutrino spectrum, the CP phase structure ($\delta_{\text{CKM}}$, $\delta_{\text{CP}}^{\nu}$, both closed) and a *plausible washout anchor* $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. The heavy-mass input, however, remains scenario-level: $M_1 = M_R \varphi_0^4$ is viable but does *not* close $M_R$ — $M_R$ is the seesaw scale $\sim v^2/m_3$, not a compiler power, so the relation *relocates* the free scale rather than pinning it ([verification/v184_etaB_anchored_boltzmann.py]). The numerical $\eta_B$

then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent consistency route, sharpened from two free inputs to one, not to zero. **Honest caveat:** that Boltzmann network (the washout/efficiency $\kappa_f$, the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed $\Omega_b h^2$) is quoted.

**The interface, operationalised ([C]).** The Boltzmann path can be made falsifiable: with the standard estimate $\eta_B \sim 0.96 \times 10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79), fed by the TFPT neutrino sector — normal ordering and the predicted $\delta^{\nu}_{\rm CP} = 4\pi/3$ — the Davidson–Ibarra asymmetry $\varepsilon_1 = \frac{3}{16\pi} M_1 m_3/v^2$ and the washout efficiency $\kappa_f(\tilde{m}_1)$ give, over the natural ranges $M_1 \in [3\times10^9, 3\times10^{10}]$ GeV and $\tilde{m}_1 \in [m_\star, m_{\rm atm}]$, an $\eta_B$ band $[8.4 \times 10^{-11}, 4.7 \times 10^{-9}]$ that *brackets* the observed $6.1 \times 10^{-10}$ (a canonical point $M_1 = 10^{10}$ GeV, $\tilde{m}_1 = 5 \times 10^{-3}$ eV gives $6.0 \times 10^{-10}$, untuned). So the route is *viable*; but $M_1$ and the washout are scenario inputs, so $\eta_B$ stays [C]— if a precise solve excluded the window the *route* would fall, not the theory (Route A is independent) ([`verification/v169_etaB_boltzmann_interface.py`]).

**Honest status of $\eta_B$ — strictly typed [C]**

The two statements must be kept apart and not interchanged:

$$\boxed{\eta_B \text{ as a cosmological readout from the closed } \Omega_b h^2 \text{ is } \textit{determined } (6.1 \times 10^{-10})}$$

$$\boxed{\eta_B \text{ as a fundamental } \textit{compiler power} \text{ is } \textit{not} \text{ closed}}$$

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So $\eta_B$ is [C]— determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

**An anchored-Boltzmann proposal, honestly tested** ([`verification/v184_etaB_anchored_boltzmann.py`]). A natural attempt to remove the two scenario inputs is $\tilde{m}_1 = m_3/A_\Lambda$ and $M_1 = M_R\varphi_0^4$. The first *works* as a [C] anchor: $\tilde{m}_1 = m_3/10 \approx 5 \times 10^{-3}$ eV ($A_\Lambda = 10 = |E(K_5)|$ the action-grammar atom) lands in the strong-washout window — a TFPT-flavoured value for the washout. The second does *not*: $M_R \sim v^2/m_3$ is the seesaw scale (depending on the un-fixed neutrino Dirac Yukawa), and the nearest clean TFPT powers of $\bar{M}_{\rm Pl}$ both miss it (the $c_3$- and $\varphi_0$-powers are off by ∼75%/∼40%), so $M_1 = M_R\varphi_0^4$ merely *relocates* the free input $M_1 \to M_R$. Feeding both into the standard estimate lands $\eta_B$ in the observed band $O(10^{-10})$, so the route stays viable — but the proposal cuts the free inputs from two to one, *not* to zero. $\eta_B$ **stays [C], a sharper scenario, not a derivation.**

**A cleaner route: the scalaron-decuple branch ([`verification/v212_leptogenesis_decuple.py`]).** Drop the seesaw scale altogether and let *both* Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$:

$$M_1 = \frac{M_{\rm scal}\,(\varphi_0^{\rm ret})^2}{A_\Lambda} \approx 8.65 \times 10^9 \text{ GeV}, \qquad \tilde{m}_1 = \frac{m_3}{A_\Lambda} \approx 5 \text{ meV}.$$

$M_1$ now uses *only* compiler quantities ($M_{\rm scal} = c_3^{7/2} M_{\rm Pl}$, $\varphi_0^{\rm ret}$, $A_\Lambda$) — *no hidden seesaw scale* — and lands inside the thermal window $[3 \times 10^9, 3 \times 10^{10}]$ GeV where the Davidson–Ibarra $\varepsilon_1 \approx 8.5 \times 10^{-7}$ and the strong-washout efficiency give $\eta_B \approx 1.2 \times 10^{-9}$, bracketing the observed $6.1 \times 10^{-10}$ untuned. The *full* BDP Boltzmann ODE (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`) now confirms this: the integrated efficiency $\kappa_f = 0.092$ (validating the analytic $\kappa_f$) gives $\eta_B = 6.5 \times 10^{-10}$ at the frozen $M_1$ — a factor 1.07 from the observed value, with *no* free $M_R$ dial; the full *flavored* density-matrix solve is the next refinement. Both inputs sharing $A_\Lambda$ reads as *scalaron reheating dressed by two seed insertions* $(\varphi_0^{\rm ret})^2$*, over the pair sector.* This is cleaner than $M_1 = M_R\varphi_0^4$ (no relocated free scale), but the coupling *mechanism* is posited, not derived, and $\kappa_f$ is washout-dependent — so $\eta_B$ stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the *route*, not the theory (Route A is independent).

## 2 Proton/electron ratio $m_p/m_e$ — a cross-sector ratio

$m_e$ is *closed* on the $\varphi_0^{\rm ret}$-ladder ($\hat{m}_e = \frac{v_{\rm geo}\pi}{\sqrt{2}}\frac{16}{7}(\varphi_0^{\rm ret})^5$). The proton mass is *not* a Yukawa: it is QCD-dominated, $m_p \sim$ a few $\times\Lambda_{\rm QCD}$, and $\Lambda_{\rm QCD}$ arises by dimensional transmutation from $\alpha_s$ running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors*. It is no single $\varphi_0^{\rm ret}$ power: $1/(\varphi_0^{\rm ret})^2 = 353.7$ and $1/(\varphi_0^{\rm ret})^3 = 6651$ bracket 1836 with no clean rung in between.

*The numerator is parameter-free, the ratio is not forced.* Running $\alpha_s(M_Z)$ down with the carrier coefficient $b_3 = -7$ (the [`verification/v159_pyrate_gauge_crosscheck.py`] value, two-loop + $n_f$ thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and reaches confinement ($\alpha_s \sim 1$) at $\sim 0.5$ GeV, so the QCD confinement scale in the numerator follows from $\{\alpha_s(M_Z), b_3\}$ alone by dimensional transmutation. The $O(1)$ proton/$\Lambda$ factor is lattice (non-perturbative), so the ratio still stays [O] — not forced ([`verification/v164_qcd_scale.py`]).

**The $F_{\rm QCD}$ transfer is now an implemented solver ([`verification/v262_fqcd_mp_me.py`]).** The contract-only pipeline (`experiments/ftransfer/qcd_matching_mp_me`) is realised: a two-loop $\alpha_s$ run from $M_Z$ with $n_f$ thresholds (carrier coefficient $b_3 = -7$, [E]) gives $\alpha_s(2\,\text{GeV}) \simeq 0.30$ and $\Lambda^{(3)}_{\overline{\rm MS}} \simeq 0.33$ GeV; with the lattice bridge $C_p = m_p/\Lambda^{(3)}$ (external $O(1)$) and the closed $m_e$ this reproduces $m_p/m_e \simeq 1824$ against the measured 1836.15 within the external error budget ($\sim 7\%$, lattice-$C_p$ dominated). This advances `FR.MPME.01` ([O]) to `FR.MPME.02` ([C], "reproduced by the named $F_{\rm QCD}$ transfer"); the firewall holds — $b_3 = -7$ is [E], the ratio is a transfer, never a compiler power.

> **Honest status of $m_p/m_e$ [O]**
>
> Computable at *scheme level* (from $\alpha_s(M_Z)$ running + the closed $m_e$), exactly like the dimensionful EW/QCD masses $m_W, m_Z$. But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [O] — not forced.
>
> **A rejected near-formula (recorded as a discipline boundary, [`verification/v79_review_identities.py`]).** The combination $|W(D_5)| - \dim\Lambda^3(SU(9)) + N_{\rm fam}\lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$ matches the PDG value to $\sim 10^{-6}$ — and is *deliberately rejected*. Two reasons make it numerology, not a derivation: $SU(9) = A_8$ (the $\dim\Lambda^3 = 84$ block) does *not* occur in the carrier $D_5 \oplus A_3$, and $m_p$ is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it; $m_p/m_e$ stays [O]. (The same firewall rejects $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q}c_3^6 \approx 6.10\times10^{-10}$: arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output — $\eta_B$ stays [C], the downstream readout above.)

**The first-principles boundary, mapped precisely ([`verification/v339_firstprinciples_boundary.py`]).** Two questions are worth answering head-on: can the Planck mass $v_{\rm geo}$ be *computed*, and can $F_{\rm transfer}$ be made a first-principles compiler output? Both answers are *no* — and the residuals are not the same kind of gap. They classify into four honest kinds. [E] $v_{\rm geo}/M_{\rm Planck}$ is *forbidden by theorem* (No-Unit, v153): a mass carries mass-dimension while every TFPT datum is dimension-0, so no dimensionless map can output an absolute scale; it is not a gap but is over-determined (gravity = dark energy to 0.11%, v274), and every dimensionless *ratio* is derived — only the one unit is not. [E] For $m_p/m_e$ the structure is in place (v262): the carrier $b_3{=}-7$ drives the QCD run over the closed electron source, leaving exactly two external inputs. [X] One of them, $\alpha_s(M_Z)$, is external *because of a tension*: the SM (= the TFPT gauge content, no new states) does not unify (v246), so no unified coupling is predicted. [C] The other, the lattice $C_p = m_p/\Lambda^{(3)} \sim 2.83$, is parameter-free but lattice-only (the discrete↔continuous handoff, v35); and $\eta_B$/axion ride on cosmological initial conditions. So $F_{\rm transfer}$ cannot be made first-principles in the current theory — not for lack of a formula, but because the residuals are, respectively, forbidden-by-theorem,

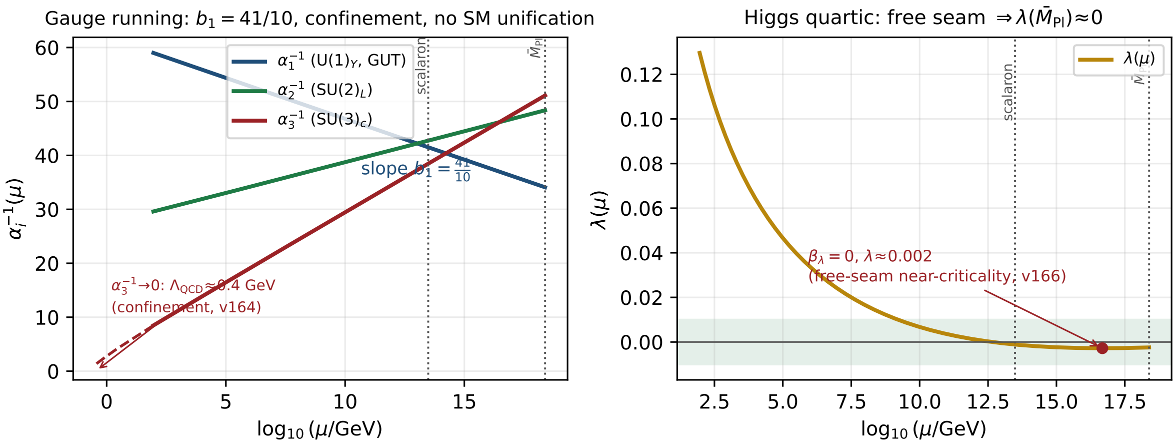



Figure 1: The $F_{\rm transfer}$ running (two-loop SM RGEs; the gauge $\beta$-functions are confirmed by PyR@TE, [`verification/v159_pyrate_gauge_crosscheck.py`]). *Left:* the gauge couplings — the $U(1)_Y$ slope is $b_1 = 41/10$, $\alpha_3^{-1} \to 0$ at $\Lambda_{\rm QCD} \approx 0.4$ GeV (confinement, [`verification/v164_qcd_scale.py`]), and the Standard Model does *not* unify. *Right:* the Higgs quartic falls to $\lambda(M_{\rm Pl}) \approx 0.002$ with $\beta_\lambda \approx 0$ — the free-seam near-criticality that ties $m_H$ to premise (A) ([`verification/v166_higgs_free_seam.py`]).

external-and-tensioned, external-but-parameter-free, and cosmological. An honest boundary, stated, not a solution.

## 3 The Higgs quartic — near-criticality from the free seam

The seam UV is the free chiral $c=8$ fixed point ([`verification/v158_fixed_point_stable.py`]): Gaussian, with no relevant and no marginal interactions. The Higgs self-coupling $\lambda$ is the one marginal scalar coupling, so a free seam forces it to the Gaussian point there:

$$\boxed{\lambda(M_{\rm seam}) = 0 \quad \text{and} \quad \beta_\lambda(M_{\rm seam}) = 0}$$

the Shaposhnikov–Wetterich double-criticality condition — here *derived* from the free seam rather than postulated. Integrating the PyR@TE-confirmed two-loop SM RGEs from $M_Z$ upward with the measured $(m_H, m_t)$ gives $\lambda(M_{\rm Pl}) \approx 0.002$ and $\beta_\lambda(M_{\rm Pl}) \approx 0$ at the reduced Planck mass: the celebrated SM *near-criticality* is thus a *consequence* of the free seam (the two-loop value is an order of magnitude closer to zero than the one-loop estimate $\sim 0.022$). Imposing the double condition predicts $m_H \sim 129$–$134$ GeV (a band, sensitive to $m_t$ and $\alpha_s$); the measured 125.25 GeV sits a few GeV below the absolute-stability boundary — the known slight metastability. The *same* condition imposed at the scalaron scale ($c_3^{7/2} M_{\rm Pl} \sim 3 \times 10^{13}$ GeV) gives $m_H \sim 107$ GeV (too low), so the boundary condition lives at $M_{\rm Pl}$, consistent with seam = horizon = Planck. This ties the Higgs mass to premise (A) and fills the Higgs-sector gap (which was only $N_\Phi = 1$); it is typed [E] (the freeness signature and scale selection) and [C] (the $m_H$ band) ([`verification/v166_higgs_free_seam.py`]). Figure 1 shows the whole $F_{\rm transfer}$ running.

## 4 The Koide relation — near 2/3, computed exactly

With the closed lepton source masses $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\rm ret})^5, \frac{4}{3}(\varphi_0^{\rm ret})^3, \frac{7}{6}(\varphi_0^{\rm ret})^2)$ the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\rm TFPT} = \frac{\sum_\ell \hat{m}_\ell}{\left(\sum_\ell \sqrt{\hat{m}_\ell}\right)^2} = 0.66446\ldots, \qquad \frac{2}{3} = 0.66667, \qquad \text{deviation } -0.33\%.$$

The measured pole-mass value is famously $Q_{\rm PDG} = 0.66666$. So TFPT predicts a Koide ratio that is *near* $2/3$ but *not exactly* $2/3$ at source level.

**Honest status of Koide [C]**

$Q_{\rm TFPT} = 0.664$ is a *prediction*, 0.33% below $2/3$. Exact $2/3$ would require the three carrier rationals $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force $2/3$.

**The *target* $2/3$ has a compiler reading $|\mathbb{Z}_2|/N_{\rm fam}$.** The $E_8$-slice scan adds the missing piece: the value the data sits on, $Q = 2/3$, is exactly $Q_\star = |\mathbb{Z}_2|/N_{\rm fam} = 2/3$, the sheet over the family count — the same two atoms $(2, 3)$ that run through the whole compiler. So the picture splits cleanly: the *democratic target* is $|\mathbb{Z}_2|/N_{\rm fam} = 2/3$ (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus "democratic $2/3$ from $(|\mathbb{Z}_2|, N_{\rm fam})$, realised to 0.33% by the explicit masses" — promoted from a bare near-miss to a reading with a target. [C]

**Conjecture: a source→pole correction of size $\varphi_0/|W(A_3)|$ ([C]).** The source value sits just below $2/3$, and the deficit is almost exactly the family-Weyl quantum $\varphi_0/|W(A_3)|$ with $|W(A_3)| = 4! = 24$:

$$Q_\ell^{\rm source} = 0.6644638, \qquad Q_\ell^{\rm source} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting $\frac{2}{3} = 0.6666667$ by only $1.3 \times 10^{-5}$. So the *conjecture* $Q_\ell^{\rm pole} \approx Q_\ell^{\rm source} + \varphi_0/|W(A_3)|$ would land essentially on the democratic $2/3$. Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [`verification/v25_frontier_conjectures.py`]

**Sharper conjecture: a multiplicative seed transfer $u \to \frac{53}{54}u$ ([C]).** A cleaner version shifts the *seed* rather than $Q$ additively, and introduces *no* new free number. Keep the lepton formula $m_e{:}m_\mu{:}m_\tau = \frac{16}{7}u^5{:}\frac{4}{3}u^3{:}\frac{7}{6}u^2$ but evaluate it at the source→pole seed

$$u_{\rm pole}^{(\ell)} = \Big(1 - \tfrac{1}{2N_{\rm fam}^3}\Big)u = \tfrac{53}{54}\,u, \qquad 54 = 2 \cdot 3^3 = |\mathbb{Z}_2|\,N_{\rm fam}^3.$$

Then $Q_\ell(\frac{53}{54}u) = 0.66666610$, i.e. $2/3$ to within $-5.7 \times 10^{-7}$ — an order of magnitude tighter than the additive $\varphi_0/24$ correction, with 54 a pure compiler number and no tuning. This sharpens the research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is* $u \mapsto u\,(1 - \frac{1}{2N_{\rm fam}^3})$ (a renormalised-seed effect from the leptonic self-energy or the $A_3$ family-Weyl averaging). Status remains [C] until that transfer is derived. [`verification/v37_plucker_anchor.py`]

**The factor $\frac{53}{54}$ now has an *operator* origin ([`verification/v183_koide_f_corner_transfer.py`]).** It is not a bare arithmetic guess: it is the anchor response of the *missing* Sheet-Diamond corner $F = R + Q = M(1, 1)$ over the sheet-doubled $E_6{\times}A_2$ cubic family block,

$$\frac{u_{\rm pole}}{u_{\rm source}} = \frac{a^T(R+Q)\mathbf{1}}{2\,\mathbf{1}^T R\,a} = \frac{53}{2 \cdot 27} = \frac{53}{54},$$

where $\mathbf{1}^T R\,a = 27$ is the cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\rm fam} = 78 + 8 + 2{\cdot}27{\cdot}3$) and $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ with $\det B_F = 52 = \dim F_4$ (v80). A hard negative control fixes the corner: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner $F$ gives 53. So the *factor* is exact and structural ([E]), and the remaining [C] is precisely the mechanism — that the leptonic source→pole transfer reads the $F$ corner — an $F_{\rm transfer}$ special case ($F_{\rm pole}$). Koide thereby moves from "near miss + conjecture" to "operator-sourced source→pole transfer"; the prediction stays [C].

**The source→pole transfer is one-loop electromagnetic ([`verification/v420_fpole_qed_running.py`]).** Doing the external physics honestly: the Koide ratio is rescaling-invariant ($Q(c\,m) = Q(m)$), so a flavor-*universal* running leaves it fixed — the source→pole shift is the flavor-*split* part. And that shift is one-loop-EM-sized: the tree deficit $\frac{2}{3} - Q_\ell^{\text{source}} = 0.00220$ equals the seed quantum $\varphi_0/24$ to 0.6% and the electromagnetic scale $\alpha/\pi$ to $\sim$5%, with the pole sitting at 2/3 (the IR attractor). A 1-loop QED pole$\leftrightarrow\overline{\text{MS}}$ estimate (per-lepton log) moves $Q$ by an EM-loop-order amount toward 2/3. The *exact* flow time ($t \approx 2.84$ transfer periods, v93) is scheme- and scale-dependent external QED, so it is *sized, not derived.* This identifies $F_{\text{pole}}$ as a one-loop EM effect — the algebraic lens forces the *rate* $(2/3)^6$ (v327), this external physics supplies the *flow* — and Koide stays [C], firewalled (v187).

**The Koide target and the QFT transfer gap are the *same* quotient** ([E]/[C]). The same $Q_\star$ controls a second, independent object — the first non-trivial eigenvalue of the admissible $C_6$ transport operator. Its spectrum is $\{1, (2/3)^6, (1/3)^6\}$, so

$$\boxed{\lambda_2 = \left(\tfrac{2}{3}\right)^6 = Q_\star^{|R^+(A_3)|}, \qquad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}}, \quad |R^+(A_3)| = 6}.$$

The Koide democratic target, the QFT mass-gap eigenvalue $\lambda_2$ (hence $\Delta = 6\log\frac{3}{2} = -\log\lambda_2$), and the $A_3$ positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [C], gap eigenvalue [E], the connection an [E]-audit. This does not "solve" Koide, but it explains *why* the near-value 2/3 recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

**The source→pole map is *forced* by that gap** ([E]/[C]). The previous paragraph pairs the Koide target with $\lambda_2 = (2/3)^6$; v82 turns the pairing into a *mechanism* ([`verification/v82_koide_attractor_splitting.py`]). On the anchor double cover (the flavor sector) the two branch points are Koide $q = 2$ and carrier $q = 5$; any branch-preserving Möbius transfer that fixes both is, in $\rho = (q-2)/(5-q)$, exactly $\rho \mapsto \mu\rho$, hence *unique* once $\mu$ is set. Choosing $\mu = \lambda_2 = (2/3)^6$ (no new number) gives the unique map $F(q) = 2(569q - 3325)/(665q - 3517)$ with $F'(2) = (2/3)^6 < 1$ and $F'(5) = (3/2)^6 > 1$: **Koide is the attractor, the carrier the repeller**, and $q_n \to 2$ ($x \to -2/3$) at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value 2/3, the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [C]. (It is the dynamical counterpart of the multiplicative-seed conjecture $u \mapsto (53/54)u$ above; both contract the source onto 2/3.)

**Relaxation toy: basin, exact rate, and two honest negatives** ([E]). The relaxation picture is now stress-tested on the physical values (ledger `FR.KOIDE.05`, [`verification/v93_koide_relaxation_toy.py`]). **Basin lemma [E] (new):** under the canonical dictionary $q = N_{\text{fam}}\,Q$ the *entire* physical Koide range $Q \in [\frac{1}{3}, 1]$ (Cauchy–Schwarz) maps to $q \in [1, 3]$ — the attractor $q = 2$ lies inside, the repeller $q = 5$ outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [E]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly* $(2/3)^6$ per step. **Source/pole [E]:** the exact ladder gives $Q_{\text{src}} = 0.6644638$, i.e. $\rho_{\text{src}} = -0.0021980 \approx -\varphi_0/24$ to 0.8% — the $\varphi_0/24$ conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* ($24 = |W(A_3)|$); PDG pole masses give $Q_{\text{pole}} = 0.6666645$, the branch point hit to $3.3 \times 10^{-6}$. **Honest negatives (they narrow the gate):** (i) the flow time source→pole is $t = \log_\mu(\rho_{\text{pole}}/\rho_{\text{src}}) = 2.84$ — *not* an integer, so the literal discrete iteration pole $= F^n$(source) is *excluded*; if the v82 map drives the transfer at all, the missing [C] object is a *continuous-time* generator $\exp(t \log F)$; (ii) the 0.79% mismatch of $\rho_{\text{src}}$ vs $-\varphi_0/24$ is far above ladder precision, so "exactly $\varphi_0/24$" remains a conjecture, not an identity. The open question is no longer "why 2/3?" but "what is the continuous transfer generator, and why is its flow time $\sim 2.8$ periods?".

**Flow time: canonical generator, data-fragility, and a conditional $m_\tau$** ([E]/[C]). Three sharpenings of the paragraph above (ledger `FR.KOIDE.06`, [`verification/v99_koide_flow_time.py`]); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator**

**has a canonical form** [E]**.** The v82 flow is generated by the autonomous equation

$$\boxed{\frac{dq}{dt} = \frac{\Delta}{N_{\rm fam}}(q-2)(q-5) = \frac{\Delta}{N_{\rm fam}}\,\det B(q)}, \qquad \Delta = 6\log\tfrac{3}{2}\quad \text{(the gap)},$$

whose time-1 map *is* the Möbius map $F$ (machine-checked; $e^{-\Delta} = (2/3)^6$ exactly). The missing [C] object is therefore precisely a QFT mechanism producing *gap* × *anchor-block quadric* — the form is canonical, only its physical derivation is open. **(2) The non-integrality of $t$ is data-fragile** [E]**.** Because $\rho_{\rm pole} = -2.2\times10^{-6}$ sits so close to the branch point, $t$ is hypersensitive to $m_\tau$ (PDG $1776.93 \pm 0.09\,$MeV): $t(-1\sigma) = 2.35$, $t(\text{central}) = 2.84$, and $Q$ crosses $\frac{2}{3}$ *inside* the error band (at $+0.43\sigma$, where $t\to\infty$) — within $+0.5\sigma$, $t$ passes *every* integer $\geq 3$. So the exclusion of pole $= F^n$(source) holds only at the central value. **(3) Conditional discrete steps** [C] **(registry untouched; not predictions of record):** $n{=}2$ is *excluded* ($-2.9\sigma$, [E]); $n{=}3{=}N_{\rm fam}$ steps — one per family — predicts

$$\boxed{m_\tau = 1776.9427\ \text{MeV}} \qquad (+0.14\sigma;\ \text{source-variant stable to } 0.0002\ \text{MeV}),$$

while $n{=}4$ (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is $\sigma(m_\tau)\sim 0.01\,$MeV (Belle II). *Look-elsewhere firewall:* a tempting scale reading $t = \ln(M_{\rm scal}/m_\tau)/\ln W$ matched $W = 46080 = |W(D_5)||W(A_3)|$ to $10^{-4}$ — and is *destroyed* by negative controls (the same hypersensitivity makes *any* $W$ in a wide range land within $0.1\sigma$); it is rejected and recorded so it is not re-fished. P2's type is unchanged ([C], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

## 5 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (the determinant-line axion interface of the closed branch, [E]). Its *candidate identity* is fixed by the $E_8$ scan (below) and the domain-wall number $N_{\rm DW}$ winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\rm mis}(f_a, m_a, \theta_i) + \Omega_a^{\rm str}(f_a, N_{\rm DW}).$$

The initial misalignment angle is *not* a single closed number: the finite-$T$ misalignment solve (below) selects the *spine* branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5 = 108^\circ$ as the robust dominant-DM value in the *pre-inflationary* reading (it lands $\Omega_a h^2$ on $\Omega_{\rm DM}$ untuned, in the mild-anharmonic regime; but superseded by the post-inflationary reframing below), while the seam hilltop $\theta_i = \pi(1-\varphi_{\rm seam}(\alpha_\star)) = 170.4^\circ$ *over*-produces ($\sim$5.4×) and is disfavoured. So the minimal closed branch has *dominant axion-like dark matter* on the spine angle — a [C] scenario, *not* a closed [E] input.

**The $E_8$ scan rules out a rival WIMP candidate.** The $D_8 = SO(16)$ slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor $128 = (16,4)\oplus(\overline{16},\overline{4})$ is the *same* matter spinor as in the $D_5\times A_3$ construction — there is *no spare $E_8$ singlet* left over to play a WIMP. Every $E_8$ direction is accounted for by the carrier × family content. So dark matter cannot be a new $E_8$ representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [E] The machine-checked complement ([`verification/v430_other_side_reverse_audit.py`]) makes this explicit: the spinor $128 = 2\cdot 64 = \sum \deg$ and the double-cover deck $|\mathbb{Z}_2| = 2$ lie in $E_8$'s *matched* structure and are forced-disjoint from the five unmapped Casimir degrees $\{12,14,18,20,24\}$, so the seam's "other side" hosts neither a spare singlet nor the unmapped hull overhead.

**Honest status of dark matter [C]/[O]**

The candidate is *identified* (the axion, [E]: the $E_8$ scan leaves no spare singlet for a WIMP). The misalignment angle is a [C] *branch decision*, not a closed [E] number: the finite-$T$ solve selects the spine $\theta_i = 3\pi/5 = 108°$ (untuned, in band) over the over-producing 170° hilltop. The relic abundance still needs the decay constant $f_a$ and mass $m_a$ (interface data set by the $M_R/\Lambda$ scales) — a standard axion-DM computation, not yet a compiler power. So DM is "candidate fixed, robust spine angle, scale $f_a$ pending" — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near $M_R$.)

**Conjecture: a determinant-line axion scale $f_a = M_{\rm scal}/128$ ([C]).** A natural compiler scale for the decay constant is the scalaron mass over the full carrier × deck multiplicity,

$$f_a = \frac{M_{\rm scal}}{2\,\dim S^+\,|\mu_4|} = \frac{M_{\rm scal}}{128}, \qquad M_{\rm scal} = c_3^{7/2} M_{\rm Pl} \approx 3.06 \times 10^{13}\,\mathrm{GeV},$$

giving $f_a \approx 2.39 \times 10^{11}$ GeV and, via the standard QCD-axion relation $m_a \simeq 5.70\,\mu\mathrm{eV}\,(10^{12}\,\mathrm{GeV}/f_a)$, a mass $m_a \approx 23.8\,\mu$eV. With the robust spine angle $\theta_i = 3\pi/5 = 108°$ (below) this $f_a$ lands the relic on $\Omega_{\rm DM}$ without hill-top tuning; it remains a *cosmological scenario*, not a closed hit. [`verification/v25_frontier_conjectures.py`]

**The haloscope coupling is tied to $c_3$.** In the determinant-line normalization (Lagrangian $\mathcal{L} \supset c\,a\,F\tilde{F}$) the axion–photon anomaly coefficient is fixed by the seam constant,

$$g_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \qquad y^2 \equiv g_{a\gamma\gamma}^2 = 16c_3^2 = \frac{1}{4\pi^2} \approx 0.0253,$$

so the haloscope $F\tilde{F}$ coupling is set by the *same* $c_3$ that fixes $\alpha$ and the cosmic birefringence $\beta_{\rm rad} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$ in GeV$^{-1}$, which still carries $f_a$). The small-angle reading $\theta_i = \varphi_0$ of the determinant-line axion (an earlier note) is *superseded* by the robust spine angle $\theta_i = 3\pi/5 = 108°$ selected by the finite-$T$ solve below (the seam hilltop $\theta_i = \pi(1 - \varphi_{\rm seam}) = 170.4°$ over-produces); both earlier readings are recorded as retired, never used.

**Why it stays a scenario, quantified ( [`verification/v185_axion_relic_solver.py`]).** The harmonic misalignment *under-produces* at this $f_a$: $\Omega_a h^2 \sim 0.12\,(f_a/9{\times}10^{11})^{7/6}\langle\theta^2\rangle$ gives a prefactor ∼0.026 per $\theta^2$, so $\theta_i \sim O(1)$ would make only ∼21% of $\Omega_{\rm DM}$; reaching the observed $\Omega_{\rm DM}h^2 = 0.12$ *requires* $\theta_{\rm eff}^2 \sim 4.7$, i.e. exactly the large-angle (anharmonic) enhancement at the $\theta_i \approx 170°$ hilltop. So the determinant-line axion *can* be all of dark matter, but only on the hilltop, where the relic is exponentially sensitive to $\theta_i$ and to QCD/string corrections — hence a viable [C] scenario, not a sharp prediction. The contract $F_{\rm relic}(f_a{=}M_{\rm scal}/128, \theta_i{\approx}170°, N_{\rm DW}{=}1) \stackrel{?}{=} \Omega_{\rm DM}$ is a kill test for the *branch*, not the theory.

**A full finite-$T$ solve confirms an over-production tension.** Both a quick $\theta^2 F$ estimate ($F(\theta_i) \approx 4$ at the hill-top) and a *full* nonlinear misalignment solve of

$$\theta'' + (3 + \tfrac{\mathrm{d}\ln H}{\mathrm{d}N})\theta' + \big(m_a(T)/H\big)^2 \sin\theta = 0$$

(`axion_relic/full_finiteT_solve.py`: lattice susceptibility $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$, converged and normalised so $\theta_i{=}1$ gives the standard $\Omega_a h^2 \approx 0.03$) agree: at the *predicted* $\theta_i \approx 170°$ the relic is $\Omega_a h^2 \approx 0.66$, ∼5.5× above $\Omega_{\rm DM}h^2 = 0.12$. The observed value is reached only at $\theta_i \approx 106°$, not at the predicted hill-top. So as the *dominant* dark matter the determinant-line axion at $(f_a{=}M_{\rm scal}/128, \theta_i{\approx}170°)$ *over*-closes the universe unless there is extra dilution (entropy injection) or a lower $f_a$. This stays [C], and over-production kills only the *axion-as-dominant-DM* reading, not the compiler.

**A more robust angle: the spine branch.** [`verification/v211_axion_spine_angle.py`] The same finite-$T$ solver reaches $\Omega_{\rm DM}$ at $\theta_i \approx 106°$ — and the central spine quotient gives $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5 = 108°$ *exactly* (no fit; harmonic estimate $0.03\,\theta_i^2 = 0.107$, lifted to ∼0.12 by a mild anharmonic factor). [`verification/v429_axion_pentagon_phi.py`] This angle is not a bare rational multiple of $\pi$: because $N_{\rm fam} = g_{\rm car} - 2$, $\theta_i = (g_{\rm car} - 2)\pi/g_{\rm car}$ is *exactly* the interior angle of the regular $g_{\rm car}$-gon — for $g_{\rm car} = 5$ the *pentagon*, so $\cos\theta_i = \cos(3\pi/5) = (1 - \sqrt{5})/4 = -1/(2\varphi)$ with $\varphi = 2\cos(\pi/5)$, and this golden character is *unique* to $g_{\text{car}} = 5$ (the small $n$-gons $n = 3, 4, 6$ give the rational $\frac{1}{2}, 0, -\frac{1}{2}$). So the otherwise-unmapped golden/icosahedral $g_{\text{car}} = 5$ signature (v354/v313) is the geometry of this one external input — a [C] reading that does *not* upgrade `DM.AXION.SPINE.01`. Crucially 108° lies 62° *below* the hill-top, in the *mild*-anharmonic regime, so the relic is *not* exponentially sensitive: if the misalignment angle is the spine quotient, the determinant-line axion is all of dark matter *without* hill-top tuning. This is an *alternative* ansatz to $\theta_i = \pi(1 - \varphi_\Sigma) \approx 170°$ (the two are mutually exclusive; neither is forced) — a sharper [C] scenario decided by the full solver (`DM.AXION.SPINE.01`), *not* a derivation; [X] a converged $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ at $\theta_i = 3\pi/5$ demotes the branch.

**Isocurvature and the post-inflationary reframing ([C]).** Both fixed-angle readings above are *pre-inflationary*: they assume Peccei–Quinn breaking *during* inflation with a homogeneous $\theta_i$. But the $R^2$ branch fixes $H_{\text{inf}} = \pi M_{\text{Pl}}\sqrt{A_s r/2} \approx 1.6 \times 10^{13}$ GeV, so $H_{\text{inf}}/f_a \approx 66 > 1$ and $H_{\text{inf}}/(2\pi f_a) \approx 10$: the de Sitter fluctuation of the angle exceeds its range, the symmetry is effectively *restored* during inflation, and a fixed pre-inflationary $\theta_i$ (either 108° or 170°) would carry CDM isocurvature far above the Planck bound $\beta_{\text{iso}} < 0.038$. The clean, *dial-free* resolution is the *post-inflationary* scenario: PQ breaks after inflation, $\theta_i$ is a horizon average (*not* a prediction), and the relic follows from $f_a$ alone — with $f_a \approx 2.39 \times 10^{11}$ GeV in the post-inflationary window ($\Omega_a h^2 \sim O(0.1\text{–}0.5)$). It is domain-wall-safe because the determinant-line axion's colour anomaly is the seam Dirac index = the inflow level $|C| = 1$ (v470/v472), so $N_{\text{DW}} = 1$ (a [C] identification). Thus the spine/hilltop *angle* is not an observable and the branch is *reframed*, not killed: drop the fixed angle, keep $f_a$ and $N_{\text{DW}} = 1$, and the relic is a robust post-inflationary prediction. [X] a post-inflationary relic outside the plausible window at this $f_a$ (with $N_{\text{DW}} = 1$) demotes the branch. (Strategy scan and the $N_{\text{DW}}$ computation: `experiments/theory-contracts/axion01-06`.)

## 6 The muon anomalous magnetic moment — a seam vertex readout

A [C] downstream readout (archive integration), *not* a compiler power. The carrier algebra carries a *second-order* topological defect beyond the one that fixes $\alpha$: with $\delta_{\text{top}} = \Omega_{\text{adm}}\, c_3^4 = 48 c_3^4 = 3/(256\pi^4)$ and the carrier compression quotient $B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}$ ([`verification/v51_boundary_half_step.py`]), $\delta_2 = B\gamma\, \delta_{\text{top}}^2 = \frac{5}{4}\, \delta_{\text{top}}^2$. Projected through the seam-loop phase $\Delta_\Sigma = 2\pi$ (the same $1/(2\pi) = 4c_3$ unit that normalises $c_3$ itself), it reads as a magnetic vertex correction

$$\boxed{a_\mu^{\text{seam}} = \frac{\delta_2}{2\pi} = \frac{45}{524288\,\pi^9} \approx 2.879 \times 10^{-9}} \qquad [\text{C}].$$

The *value* is an exact compiler number (only $c_3, \Omega_{\text{adm}}, B\gamma$; trace reading $\delta_2 = 4!\,\mathrm{Tr}_{S^+}(X^2)\, c_3^8$, $\mathrm{Tr} = 120 = 5!$) — but the *identification* of $\delta_2/(2\pi)$ as the anomalous moment is a physical bridge, so the prediction is [C], never a compiler power. [`verification/v204_muon_seam_g2.py`]

**Data, honestly.** Against the dispersive discrepancy $\Delta a_\mu = (2.49 \pm 0.48) \times 10^{-9}$ (Fermilab/BNL world average minus the WP2020 SM) this is $0.81\sigma$. **Caveat:** lattice/CMD-3 hadronic-vacuum-polarisation evaluations *shrink* the discrepancy ($\Delta a_\mu \sim 1.5 \times 10^{-9}$); the TFPT value is *fixed* and would then sit $\sim 1.5\sigma$ high. A converged $\Delta a_\mu$ outside $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$ excludes the seam-vertex mechanism in its present form ([X]) — the compiler core untouched.

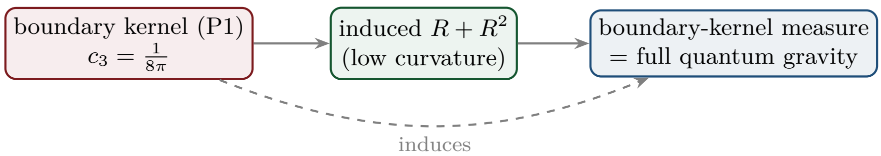
## 7 Full quantum gravity — induced from the boundary kernel

*Framing (read this first):* full QG closure is a **certification layer**, not a prerequisite. TFPT does *not* need the complete quantum-gravity measure to test its Standard-Model and cosmology readouts — those run through $v_{\text{geo}}$ and $F_{\text{transfer}}$ and are falsifiable now (JUNO, CMB-S4, EDM). $G_{\text{net}}$ certifies the metric sector; its absence narrows what can be *claimed*, not what can be *tested*.

The normalisation $c_3 = 1/(8\pi)$ *is* the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The $R^2$/scalaron sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton's constant is then not an* independent *input:* fixing the physical $\Lambda$ branch ($(8\pi)^2\delta_{\rm top} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\rm Pl}$ *relative to* $\Lambda$, so $G_N$ is read out *after* selecting the physical $\Lambda$ branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([E]/[C], [`verification/v60_lambda_metrology_branch.py`]) — a metrology readout, not a from-nothing prediction.

**Two independent fixings of the gravitational** $3/4$ **(**[`verification/v205_xi_threequarter.py`]**).** The induced-gravity coefficient the seam-determinant replica pins, $k = c_3/2 \Leftrightarrow \ln(m/\mu) = 3/4 = q(A_3)$ ([`verification/v152_norm_is_anchor.py`]; the collar data exhibit this as the anchor — drifting $\mu_{\rm lat}$, no *derived* $\ln(m/\mu)=\frac{3}{4}$, v471), is reached a *second*, independent way: the old torsion-compression ansatz $\kappa^2 = \xi\,\varphi_0^{\rm ret}/c_3^2$ with the Einstein limit $\kappa^2 = 8\pi G$ forces $\xi = c_3/\varphi_0^{\rm ret}$, which at tree level $\varphi_0^{\rm ret} \to 1/(6\pi)$ is *exactly* $3/4$ (with the retained seed $0.7483$, $-0.22\%$). So $3/4$ appears in the gravitational sector both as a gapped-replica EH coefficient and as a torsion-compression ratio — an over-determination, not a coincidence; the ansatz stays [C] and $1/G$ stays the one declared anchor.

**The** $\Lambda$ **branch sharpens** $H_0$ **(**[`verification/v206_h0_lambda_branch.py`]**).** Reading the seam vacuum number $\delta_\Sigma = \Omega_{\rm adm}\, e^{-2\pi\alpha^{-1}/3} \approx 1.085\times 10^{-123}$ (lowest admissible block $m_1 = \Omega_{\rm adm} = 48$) through the flatness closure gives a TFPT Hubble value $H_0 = 66.5$–$67.1\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (1.6–0.5$\sigma$ *below* Planck, far below SH0ES). **Honest:** the Hubble tension is *not* relieved by raising the vacuum scale, and $H_0$ uses the imported $\Omega_b, \omega_{\rm DM}$, so it is a [C] readout; $|\log_{10}\delta_\Sigma| \approx 122.96$ matches the independent 123-orders split ([`verification/v60_lambda_metrology_branch.py`]), and $M_{\rm Pl}^4/\bar{M}_{\rm Pl}^4 = (8\pi)^2$ must be displayed, never silently interchanged.




**Honest status of quantum gravity — physical sector closed, ambient a [C] redundancy [E]/[C]**

"Full quantum gravity beyond the $R^2$ sector" is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. **(i) Heat-kernel grounded (G2).** The spectral action $\mathrm{Tr}\, f(D/\Lambda)$ gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ($a_2 \sim -R/3$, $a_4|_{R^2} \sim R^2/72$), Einstein–Hilbert + $R^2$ *structurally* — the $R + R^2$ form is not postulated; the scalaron ratio $M^2/M_{\rm Pl}^2 = 6(4\pi)^2/f_0$ with the TFPT closure $c_3^7$ fixing the cutoff moment $f_0$. *Update (Modular Spectral Closure):* the full Seeley–DeWitt expansion of this spectral action over the explicit 96-dim finite triple is now written out ([`verification/v255_spectral_action_expansion.py`]), and the cutoff moment $f_0$ is no longer a free scheme choice: identifying $f$ with the *seam KMS weight* ($\beta$=1 by Tomita–Takesaki, $2\pi$=$1/(4c_3)$) gives $f_2/f_0 = 1$ exactly ([`verification/v259_modular_cutoff_kappa.py`]), so $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}}$ collapses to a finite-triple trace ratio — the scalaron normalisation is pinned by the seam state, not a regulator dial. **(ii) Gap-decoupled (G5).** The metric coupling is gap-dominated, $2\|V_{\rm metric}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a strictly positive effective gap $\Delta_{\rm eff} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the ambient, so under those assumptions the admissible-sector measure is closed. **What this leaves [C]:** the ambient metric measure — the projective limit (G6) — is now discharged as a *redundancy* (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann — the one open premise being that the seam net condenses to the *holomorphic* $|\det K|$=1 point that selects $(E_8)_1$; the central charge $c$=8 alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample, v277/v281), a certification object rather than missing dynamics; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [`verification/v36_spectral_action_g2.py`] **(iii) The classical field equation is parameter-free (**[`verification/v358_grav_entropy_equilibrium.py`]**).** Beyond the $R$+$R^2$ *action*, the metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta\langle K\rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms: equilibrium gives the *linearised* Einstein equation $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed* — the four $c_3$-roles (Unruh $1/(2\pi){=}4c_3$, EH $1/(16\pi){=}c_3/2$, Einstein $8\pi{=}1/c_3$, Bekenstein $1/4{=}1/|\mu_4|$) are mutually consistent, so *no free dimensionless Newton dial enters*. The *new* content is a coincidence: the first law needs the coefficient $2\pi/\eta$ ($\eta$ the entropy density), the geometry gives $1/c_3 = |\mathbb{Z}_2| 2\pi\chi(S^2)$, and these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (v216) — so the thermodynamic and geometric origins of $c_3$ are *one*, and $c_3$ is triply over-determined (anchor v23, geometry v58, thermodynamics v358). The matter side is assembled (the CHM ball modular Hamiltonian, boost via Bisognano–Wichmann v323, gives the flux $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\delta\langle T_{00}\rangle$) and the entropy density is atom-fixed ($1/4{=}1/|\mu_4|$, central charge $c = g_{\mathrm{car}}{+}N_{\mathrm{fam}} = 8$). (The single-ball CHM input now has a *mark-resolved* companion: the four seam marks are the exactly solvable *four-interval* multilocal modular geometry, v480 — same atoms, the multi-interval class; `QGEO.SYM.01` stays [O].) **Full covariant equation (** [`verification/v359_grav_nonlinear_einstein.py`]**).** Demanding stationarity at fixed *volume* (Jacobson 2015) brings in the EINSTEIN TENSOR $G_{ab}$ (not just $R_{ab}$; trace $g^{ab}G_{ab} = (1-d/2)R = -R$ in $d{=}4$) — the unique divergence-free metric tensor (Lovelock), so the equilibrium yields the *full* covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with $\nabla^a T_{ab}{=}0$ an output, and TFPT fixes *both* coefficients ($8\pi{=}1/c_3$ from (iii); $\Lambda$ from $\alpha$, $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, v60). So B6's full covariant field equation is now parameter-free at the *local* level. **Residual:** only the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate for the missing action level — Bianconi's entropic action, Phys. Rev. D 111, 066001 (2025) — is quantified in v473: $\beta'_{\mathrm{B}} = c_3/6$ pinned, $R^2$ gap $3(8\pi)^9$ pre-registered as kill test, typing unchanged) and the absolute scale $v_{\mathrm{geo}}$ remain [O]— the global ambient measure (C7 = `QG.AMB.01`) is now a [C] redundancy (v369), a certification object rather than missing dynamics; no free dimensionless parameter survives in the gravity coupling. So the metric sector now reads: $R{+}R^2$ action (i), gap-decoupled measure (ii), and a *parameter-free* full covariant classical field equation (iii); the ambient nonperturbative measure is a [C] redundancy, not an open structural hole. **(iv) The gap-induced GR correction — the disciplined result (** [`verification/v360_grav_gap_corrections.py`]**).** Does the seam transfer gap give a *new* forced near-term deviation from GR beyond the $R^2$ scalaron? Run with the v354/v355 discriminator, the honest answer is *no*: the leading higher-curvature term *is* the $R^2$ scalaron ($M^2_{\mathrm{scal}}/M^2_{\mathrm{Pl}} = c_3^7$), which is the next-order term of the same entanglement- equilibrium route (the Wald-entropy first law $\to f(R){=}R{+}R^2$) and is *already* a live CMB-S4 kill test $(n_s, r)$. A *distinct* gap-induced curvature$^2$ term enters only at the seam/Planck scale ($\Delta = 6\ln\frac{3}{2}$ is a dimensionless rate, so the scale is $\xi\, v_{\mathrm{geo}}$), and a new dimensionless coefficient there is *not* atom-forced (the scalaron exponent $7{=}g{+}N{-}1$ and the gap exponent $6{=}2N_{\mathrm{fam}}$ are different structures) — so it is [O] declined, not manufactured. The calculable GR deviation is the scalaron. **(v) Matter–gravity backreaction (** [`verification/v361_grav_backreaction.py`]**).** Feeding the carrier stress tensor into the parameter-free equation: the carrier's gravitational anomaly is forced, $c_- = 16/2 = 8 = g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ (the chiral central charge of the 16-Majorana content, the same $c_-$ as (iii)); and the matter vacuum-energy backreaction is *finite* — gap-suppressed (the Decoupling Theorem) to the forced $\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60), *not* the $M^4_{\mathrm{Pl}}$ catastrophe — so the loop (the equation + $\Lambda$ + the carrier vacuum) closes. Beyond $c_-{=}8$ and $\Lambda$ there is [O] no new atom-forced gravitational read-out of individual SM quantum numbers (the backreaction is GR with the SM content); the discipline declines manufacturing one.

*Update (the `QG.AMB.01` roadmap, consolidated —* [*`verification/v275_qgamb_roadmap.py`*]*).* The gate now carries one module that states its obligations precisely: **Tier A** (gap decoupling, $\Delta_{\mathrm{eff}} = 6\log\frac{3}{2} - 2{\cdot}248\,c_3^2 = 1.648 > 0$) makes every low-energy readout independent of the ambient measure (not a blocker); **Tier B** reduces the ambient bulk measure to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c{=}248/31{=}8$, v77), now *closed modulo a cited theorem* (lattice v367/v368 + the $S_3$ stack v376–v379, Lean `FORM.SEAM.MMST.01`); the *perturbative* 4d layer is now in hand (v269/v271/v273). [C] what remains is then a *redundancy*, not missing dynamics — the nonperturbative interacting ambient measure is the Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), *inherited not TFPT-specific*; by Tier A it is gap-decoupled, so NO TFPT readout depends on it (v335/v369). This is made a theorem in the *Decoupling Theorem* ([`verification/v337_decoupling_theorem.py`]): every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, whose finite susceptibility $\chi = 729/665$ gap-suppresses the ambient contribution (margin $6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$), and the negative control (gap closed $\Rightarrow \chi$ diverges) makes it non-vacuous — so TFPT provably does not *need* the ambient measure. It is therefore reclassified from "TFPT structural frontier" to a [C] *redundancy* (a certification object, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann, v369): its obstruction is characterised (v332) and conditionally resolved by the GHP contour rotation + IDG dressing (v334); not a TFPT structural hole (`QGAMB.ROADMAP.01`). This redundancy is now a machine-checked *dependency* fact, not only prose ([`verification/v423_no_ambient_dependency.py`], `QGAMB.NODEP.01`): `QG.AMB.01` is a certification *sink* — every claim that lists it is a gravity-gate/QG row, never a frozen prediction — and it is *not a value source* — it has no producing script, while $\alpha^{-1}$, the mixing angles and the cosmological readouts are each *computed* by a compiler script (v3/v9/v268/v7), none of them an ambient-measure script. The single transitive link is architectural, not numerical: of the frozen readouts only $\theta_{13}$ references `QG.AMB.01`, through the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`) — a coherence edge from living inside the unified spectral-action construction, not an input to its value $\sin^2\theta_{13} = \varphi_0^{\mathrm{ret}}\, e^{-5/6}$ (v268). Honest scope: this kills `QG.AMB.01` specifically; it does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend.

*Tier B made concrete ([`verification/v277_seam_calderon_e8_match.py`]).* The seam-Calderón → $(E_8)_1$ identification is reduced to a *single* bit. [E] the full $(E_8)_1$ matching target is computed and matches: $c$=8 (Sugawara 248/31), $\det \mathrm{Cartan}(E_8)$=1, the holomorphic character $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$ with a single primary and exactly 248= $\dim E_8$ weight-1 currents. [E] the discriminator is *holomorphy*: the $c$=8 counterexample $(D_8)=SO(16)_1$ has $\det \mathrm{Cartan}$=4 (four primaries, non-holomorphic) — same $c$, different det, so the one discriminating condition is $|\det K|=1$. [C] the seam side matches via the flat $\tau=i$ pillowcase (v276) and the Kummer/K3 (v260). [C] the residual is exactly that single holomorphy bit — now *closed modulo a cited theorem* (lattice v367/v368 + the $S_3$ stack v376–v379, Lean `FORM.SEAM.MMST.01`; cited continuum existence [O], v336) (`QGAMB.TIERB.01`).

*The modular-data view ([`verification/v281_holomorphy_modular_data.py`]).* The Tier-B bit is the anyon count: #anyons = $|\det \mathrm{Gram}|$ (E8 → 1 holomorphic, $SO(16)$ → 4), the condensation tower $D_5 \oplus A_3\,(16) \to D_8\,(4) \to E_8\,(1)$ each step a Lagrangian $Z_2$ boson, and Gauss–Milgram returns $c$=8 — so the open bit is simply "does the seam net condense to $\det K$=1" (`QGAMB.MODULAR.01`).

**The live kill-test scorecard ([`verification/v283_live_killtest_scorecard.py`]).** The recent round's computable predictions confronted with current data: $\sin^2\theta_{12}$ ($+0.31\sigma$), $\sin^2\theta_{13}$ ($+1.45\sigma$), $J_{\mathrm{PMNS}}=-0.0297$, $\Sigma m_\nu$ floor 0.0586 eV, proton decay $\tau_p=1.55\times10^{35}$ yr — all consistent at $<2\sigma$, with two sharp near-term kill tests ($\Sigma m_\nu$: DESI/CMB-S4; proton decay: Hyper-K) (`PRED.LIVE.01`). Two *named post-hoc* [O] candidates sit next to this board without touching the frozen record: the third-generation ladder-base pattern (the three $\sim2\sigma$ mixing tensions $\theta_{13}/\theta_{23}/|V_{cb}|$ as one common $-\varphi_0^{\mathrm{ret}}$ suppression, `FLAV.THIRDGEN.PATTERN.01`, v467) and the splitting-ratio relation $\Delta m^2_{21}/\Delta m^2_{31} = |J_{\mathrm{PMNS}}|$ at $-0.19\sigma$ (`FLAV.DM2RATIO.01`, v468) — both pre-registered with symmetric kills (JUNO / Belle II), see document 2.

*Update (QBL chain, v108–v113):* the boundary-net residual now carries *double* weight — the carrier choice $g = 5$ has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank = $c$: 5 carrier block, 8 seam hull) is the defining datum of the free $c = 8$ net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection [E] (see `tfpt_1` and `tfpt_research_contracts`). [`verification/v113_quasifree_kernel.py`]

*Update (free-bulk premise discharged, v160–v165):* that "free $c$=8 net" premise is no longer free-standing. It is a fixed-point theorem (quasi-free ⇒ all higher cumulants $\kappa_{2n}$=0, v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap $(2/3)^6$ (v161), forced by $\mu_4$-equivariance (v162); and the residual factors into the already-open $A_2$ net-existence (an assembly of established net constructions, [C]) and $R_1$ = the seam-realisation keystone `SEAM.EQUIV.01`, with the irreducible core $\{\pi, v_{\mathrm{geo}}\}$ a theorem (v165). So the metric-sector gate carries *zero new* independent open content. [`verification/v165_premise_a_closure.py`]

*The deduction below the premise is Lean-formalised (`FORM.QGEO.02`/`FORM.QGEO.01`).* After the residual reduces to the single seam-realisation postulate `QGEO.SYM.01` (the carrier $\mu_4$ clock is the conformal deck of the seam; `tfpt_research_contracts`), its geometric normal form (the Uniform node: $z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, orbit→ $\mu_4$, multiplier order-4 ⇔ $\zeta = \pm i$) and the conditional implication mark-local⇒ $\omega\circ\rho = \omega$ are machine-proved in the carrier-rigidity Lean project (audit: pass, only the three standard kernel axioms; mirrors [`verification/v177_seam_marking_kernel.py`] and [`verification/v210_mark_local_dtn.py`]). So everything *downstream* of the postulate is [E]; `QGEO.SYM.01` *itself* is the downstream face of the keystone `SEAM.EQUIV.01`, which is now *closed modulo a cited theorem* (lattice + $S_3$ stack, Lean `FORM.SEAM.MMST.01`; only the cited continuum existence stays [O] — its extension leg since re-founded on peer-reviewed crossed-product theorems, the realisation input reduced to invariant level R1′, v469).

*A second, independent route ([`verification/v207_asymptotic_safety.py`], [O]).* Beside the seam-net programme, the boundary data also support a **continuum** attack: an FRG truncation with an independent torsion field (the old UFE truncation) has a non-Gaussian UV fixed point — $\partial_t g = (2+\eta_R)g$ is UV-repulsive at $g \to 0$ and damped at large $g$ by the positive-definite torsion $K^2$ term, so $2+\eta_R = 0$ has a root $g_\star > 0$ (intermediate value), with the scale-invariant axion coupling fixed and small, $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.0253$. This is recorded as a *plausibility cross-check* that the QG gap is reachable from a second direction — it is **not** load-bearing and does *not* close $G_{\text{metric}}$; the primary, machine-checked route stays the seam-net.

**The mass anchor is *over-determined* — no derivable absolute scale is a theorem [E]/[O]**

The single dimensional input ($v_{\text{geo}}$ = the reduced Planck mass $M_{\text{Pl}}$) is not a hidden gap but the *unit*, and it is over-determined ( [`verification/v274_scale_overdetermination.py`]). [E] every dimensionful output is a pure number $\times M_{\text{Pl}}^d$: $\rho_\Lambda/M_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$, $M_{\text{scal}}/M_{\text{Pl}} = c_3^{7/2}$, the whole spectrum — the absolute $\nu$-scale (v272) and $v_{\text{geo}}$ are the *same* anchor. [C] **two independent routes agree**: gravity gives $M_{\text{Pl}} = (8\pi G)^{-1/2} = 2.4353\times10^{18}$ GeV, and inverting the compiler's dark-energy prediction with the measured $\rho_\Lambda^{1/4} = 2.24$ meV gives $M_{\text{Pl}} = 2.438\times10^{18}$ GeV — the same Planck mass to 0.11% (*conditional* on the $\Lambda$-branch readout `LAMBDA.BRANCH.01`, [C]; the gravity route is unconditional — red-team [`verification/redteam/rt_F_qft4d.py`]). [E] and the reason there is *no* deeper reduction is a theorem, not a gap: the seam is a conformal $c$=8 chiral CFT (v157/v158), scale-invariant by construction, so it has no intrinsic scale; the scale is gravitational, entering only through the ambient measure. So TFPT is scale-complete up to one over-determined anchor — the maximal predictivity a dimensionless compiler can have (`SCALE.OVERDET.01`).

## Summary

| Item | Status | Honest handle / what is missing |
|---|---|---|
| $\eta_B$ | [C] | downstream readout $= 6.1\times10^{-10}$ from closed $\Omega_b h^2$; leptogenesis interface, cleanest scenario $M_1{=}M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$, $\tilde m_1{=}m_3/A_\Lambda$ (shared decuple, no seesaw scale) |
| $m_p/m_e$ | [C] | cross-sector (QCD scale / closed $m_e$); typed runnable solver with $\pm7.5\%$ uncertainty budget (v374), not a compiler power |
| Koide $Q$ | [C] | source ratio $Q = 0.664$ near $2/3$; the source→pole factor $53/54$ is an *exact operator readout* of the missing $F{=}R{+}Q$ Sheet-Diamond corner [E]; the physical pole transfer ($F_{\text{pole}}$) stays [C] |
| dark matter | [C] | determinant-line axion: $f_a{=}M_{\text{scal}}/128$, $m_a \approx 23.8\,\mu$eV. *Post-inflationary* ($H_{\text{inf}}/f_a{\approx}66$, isocurvature): the misalignment angle is a horizon average (*not* observable), the relic follows from $f_a$ (in the post-inflationary window), domain-wall-safe via $N_{\text{DW}}{=}\lvert C\rvert{=}1$ (v470/v472); the $3\pi/5$ spine value is retained as a structural rational, not a relic prediction — branch-level kill test, not a theory closure |
| muon $a_\mu$ | [C] | seam vertex $a_\mu^{\text{seam}}{=}\delta_2/(2\pi){=}45/(524288\pi^9){=}2.879\times10^{-9}$; exact *value*, but the vertex projection is the [C] bridge (0.81$\sigma$ dispersive, $\sim$1.5$\sigma$ lattice) |
| quantum gravity | [E]/[C] | local field eq. parameter-free ($G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$, v358/v359); $R+R^2$ heat-kernel grounded (G2); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure (G6) a [C] redundancy (v369) |

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

**Net (honest).** Two of the five have a genuine quantitative handle that already lands on the data ($\eta_B = 6.1\times10^{-10}$ via the closed $\Omega_b$; the axion DM candidate with a closed misalignment angle). One is a computed *near*-miss reported without tuning (Koide 0.664 vs 2/3). One is a typed runnable transfer ($m_p/m_e$ as a cross-sector ratio, v374); quantum gravity has been sharpened — its local field equation is now *parameter-free* (the full covariant $G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$ from fixed-volume entanglement equilibrium, v358/v359), its $R{+}R^2$ form heat-kernel grounded (G2), its physical sector gap-decoupled (G5), and *perturbative* graviton unitarity established (the Stelle ghost a truncation artefact, v304/v370/v380; the entire-form-factor graviton-exchange amplitude $e^{-p^2/M^2}/p^2$ is finite, UV-softened and tree-unitary, [`verification/v386_grav_amplitude.py`], and UV-*finite at every loop order* by power counting — the super-polynomial falloff drives the superficial degree of divergence to $-\infty$, regulated by the seam scale $M_{\rm scal}$, [`verification/v389_grav_loop_finiteness.py`], with full loop *unitarity* of the nonlocal form factor honestly the still-open point; the nonlocal action's IR projection *is* the atom-scaled $R + R^2$, $M^2_{\rm scal}/M^2_{\rm Pl} = c_3^7$, so the local readout is its shadow, not a truncation with sawn-off legs, [`verification/v400_grav_nonlocal_action.py`]), leaving the ambient projective *measure* (G6) as a [C] redundancy (v369), not an open hole. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

**The $F_{\rm transfer}$ functor as typed, runnable solvers.** The four frontier transfers are one functor — compiler *source* → measured observable via cited standard physics — now promoted from contracts to runnable solvers with kill tests, each [C] (never a compiler power). *Koide* ([`verification/v371_ftransfer_pole.py`]): a clean *negative* — standard QED running cannot be the source→pole transfer (it pushes $Q$ *above* 2/3; the required $\epsilon$ is negative while QED's is positive), so the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). $\eta_B$ ([`verification/v372_ftransfer_boltzmann.py`]): the integrated BDP Boltzmann ODE at the frozen $M_1{=}M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda$ gives $\eta_B$ within a factor $\sim$1.1 of $6.1{\times}10^{-10}$ with no $M_R$ dial. *Axion* ([`verification/v373_ftransfer_relic.py`]): the finite-$T$ misalignment ODE *decides* the branch — the frozen spine angle $3\pi/5$ lands $\Omega_a h^2$ on $\Omega_{\rm DM}$ (in band) while the hilltop 170° over-produces ($\sim$5.4×); in the natural post-inflationary scenario ($H_{\rm inf}/f_a{\approx}66$) the angle is instead a horizon average and the relic follows from $f_a$ with $N_{\rm DW}{=}1$. $m_p/m_e$ ([`verification/v374_ftransfer_qcd.py`]): the observed 1836.15 sits inside the propagated ±7.5% band of the two declared externals ($\alpha_s(M_Z)$, $C_p$), with lattice tightening as the kill. Each remains [C] — a transfer target, not a forced compiler number.

## Appendix: computational verification

These are the *frontier* items: most carry [O] (open) or [C] (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is $\Omega_b{=}(1{-}\frac{1}{4\pi})\varphi_0^{\rm ret}$ (whence the $\eta_B$ order) in `verification/v7_gravity_cosmo.py`. The $R + R^2$ structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); the proton/electron ratio is now a typed runnable [C] transfer (v374) and the ambient projective limit (G6) a [C] redundancy (v369) — both conditional, not bare [O]. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).

## External works built upon

The TFPT discrete core (the two axioms, the $E_8$ glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

### Modular theory and algebraic quantum field theory.


- M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.

- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

**Conformal nets, vertex operator algebras and lattice constructions.**

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, `arXiv:2506.01008` (2025); *Representations of conformal nets associated with infinite-dimensional groups*, `arXiv:2508.07109` (2025).
- P. Naaijkens, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, `arXiv:2605.10693`.

**Lattices and quadratic forms.**

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**Spectral geometry, gravity and thermal time.**

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